

Model Predictive Control

MPC Core Idea

Repeatedly solve an optimization problem over a finite horizon, execute first control, replan at next time step with updated state (and environment) information

Why?

- Intractable over long horizon
- Model mismatch/uncertain dynamics. Prediction far in the future is not reliable
- Solving Bellman/HJB is intractable (nonlinear dynamics, non-convex cost, high dimensionality)
- ...

Receding horizon planning / MPC

1. Plan a trajectory over a **finite horizon** (i.e., a sequence of control inputs)
 2. Execute the **first control input** in the sequence
 3. Go to step 1.
- Solve long horizon problem with a sequence of shorter horizon problem.
 - Just make T smaller...right?
 - **Potential issue?**

Potential challenges

- Terminal cost (e.g., reach goal) is not useful if horizon isn't long enough to reach the goal.
- **Persistent feasibility.** Will future MPC iterations be feasible?
- **Closed-loop stability.** Will the system eventually converge to the desired equilibrium point?

Need some “foresight” to account for future iterations and future state

- Let N be the shorter planning horizon
- Let t be the current time.
- We are planning from t to $t + N$

$$\min_{u=(u_{t|t}, \dots, u_{t+N|t})} \sum_{k=0}^{N-1} g(x_{t+k|t}, u_{t+k|t}) + p(x_{t+N|t})$$

Cost-to-go

$$\text{s. t. } x_{t+k+1|t} = f(x_{t+k|t}, u_{t+k|t})$$

$$x_{t+k|t} \in X, u_{t+k|t} \in U \quad \forall k$$

$$\begin{aligned} &x(t) = x_{t|t} \\ &x_{t+N|t} \in X_f \end{aligned}$$

Terminal set
constraint

Something to ensure that
the system will end up in
a “reasonable” state

Persistent feasibility theorem

- If X_f is a *control invariant set* for the system $x_{k+1} = f(x_k, u_k)$ then the MPC law is persistently feasible. i.e., the next planning step will have a feasible solution

$$\begin{aligned} \min_{u=(u_{t|t}, \dots, u_{t+N|t})} & \sum_{k=0}^{N-1} g(x_{t+k|t}, u_{t+k|t}) + p(x_{t+N|t}) \\ \text{s. t. } & x_{t+k+1|t} = f(x_{t+k|t}, u_{t+k|t}) \\ & x_{t+k|t} \in X, u_{t+k|t} \in U \quad \forall k \\ & x(t) = x_{t|t} \\ & x_{t+N|t} \in X_f \text{ Control invariant set} \end{aligned}$$

Proof (sketch)

- Suppose at time t , you have a feasible solution to the current planning problem. $\mathbf{u}_0^* = (u_0, u_1, \dots, u_{N-1})$ and $x_N \in X_f$
- At the next planning step (shift time step forward by one), note that $\mathbf{u}_1 = (u_1, u_2, \dots, u_{N-1}, \tilde{u})$ is a feasible solution to the next planning problem where $\tilde{x}_{N+1} = f(x_N, \tilde{u}) \in X_f$ ($\tilde{u}$ is a feasible control corresponding to the control invariant set definition)
- We have just found a feasible solution to the next planning problem. So we will do at least as good, if not better at the next time step.

Practical considerations

- The terminal set X_f is introduced artificially for the sole purpose of leading to a sufficient condition for persistent feasibility
- Want it to be large so it does not compromise on performance
- How do we pick X_f ?
- These sets can be computed by using the MPT toolbox
<https://www.mpt3.org/>

MPC stability theorem

Assume:

1. X, U contains the origin in their interior
2. $x = 0, u = 0$ is an equilibrium $f(0,0) = 0$
3. $\beta^- \|x\| \leq g(x, u) \leq \beta^+ \|x\| \quad \forall x \in X, u \in U$
4. $X_f \subseteq X$ is control invariant
5. The terminal cost p is a local control Lyapunov function satisfying
 - $p(x) \leq \beta \|x\|$
 - $\exists \mu \quad p(x) - p(f(x, \mu(x))) \geq g(x, \mu(x)). \quad \forall x \in X_f$

Then the origin of the closed-loop system under MPC control is exponentially stable with region of attraction X_0 (N -step BRS of X_f)

Proof (sketch)

- Show that the cost is Lyapunov stable for the closed-loop system
 - i.e., the cost is shrinking over time
- Suppose you have an optimal trajectory $\mathbf{u}_0^* = (u_0, u_1, \dots, u_{N-1})$
- At the next time step $\mathbf{u}_1 = (u_1, u_2, \dots, u_{N-1}, \tilde{u})$ ($\tilde{u}$ keep $x_{N+1} \in X_f$)
- But $\mathbf{u}_1$ is not necessarily optimal

Rough proof continued

$$\begin{aligned} J_1^*(x_1) &\leq \sum_{k=1}^N g(x_k, u_k) + p(x_{N+1}) \\ &= \sum_{k=1}^{N-1} g(x_k, u_k) + g(x_N, \tilde{u}) + p(x_{N+1}) \\ &= \sum_{k=1}^{N-1} g(x_k, u_k) + g(x_N, \tilde{u}) + p(x_{N+1}) + p(x_N) - p(x_N) + g(x_0, u_0) - g(x_0, u_0) \end{aligned}$$

$$J_1^*(x_1) \leq J_0^*(x_0) + g(x_N, \tilde{u}) + p(x_{N+1}) - p(x_N) - g(x_0, u_0)$$

$$J_0^*(x_0) - J_1^*(x_1) \geq g(x_0, u_0) \geq \beta^- \|x_0\| \quad \text{Cost is decreasing!}$$

MPC stability theorem takeaway

- If we choose the cost-to-go to be a control Lyapunov function, then we can ensure stability (with some assumptions about the problem structure)

$$\begin{aligned} \min_{u=(u_{t|t}, \dots, u_{t+N|t})} & \sum_{k=0}^{N-1} g(x_{t+k|t}, u_{t+k|t}) + p(x_{t+N|t}) \\ \text{s. t. } & x_{t+k+1|t} = f(x_{t+k|t}, u_{t+k|t}) \\ & x_{t+k|t} \in X, u_{t+k|t} \in U \quad \forall k \\ & x(t) = x_{t|t} \\ & x_{t+N|t} \in X_f \text{ Control invariant set} \end{aligned}$$

Control
Lyapunov
function

What are reasonable choices of X_f and p ?

Assuming linear dynamics & quadratic cost

Case 1: A is asymptotically stable

- $u = 0$ is a feasible control
- X_f : maximally positive invariant set for $x_{t+1} = Ax_t$
- p : Solution to the Lyapunov equation
$$-P + Q + A^T P A = 0$$

“Once you reach the end, just do nothing and you will asymptotically converge”.

What are reasonable choices of X_f and p ?

Assuming linear dynamics & quadratic cost

Case 2: A is open-loop unstable

- Let K_∞ be the optimal gain for infinite horizon LQR
- X_f : maximally positive invariant set for $x_{t+1} = (A + BK_\infty)x_t$
- p : Solution to the discrete Riccati equation
$$-P + Q + A^T P A - (A^T P B)(R + B^T P B)^{-1}(B^T P A) = 0$$

“Once you reach the end, apply the infinite horizon LQR controller”.

Some comments

- Presented cases are suboptimal choices
- Need to make sure the control constraints are satisfied (increasing R)
- At present there is no other technique than MPC to design controllers for general large linear multivariable systems with input and output constraints with a stability guarantee
- Design approach (for squared 2-norm cost):
 - Choose horizon length N and the control invariant target set X_f
 - Control invariant target set X_f should be as large as possible for performance
 - Choose the parameters Q and R freely to affect the control performance
 - Adjust p as per the stability theorem
 - Useful toolbox (MATLAB): <https://www.mpt3.org/>
- In practice, sometimes choosing a good terminal cost is enough (i.e., don't need to enforce a terminal control invariant condition), though you may be sacrificing guarantees

So far...

- Assumed perfect dynamics model
- Assumed no stochasticity
- What about moving obstacles? (Prediction models, human-robot interactions)
- Theorems hold in general but challenging for non-LQR-like structure (i.e., not linear & not quadratic)
- But provide conceptual understanding to what heuristics we would like to design for nonlinear systems
- There is a whole research area on *robust* MPC and *tube* MPC (beyond the scope of this course)

Code demo

Planning a trajectory to
avoid moving asteroids

